Research on UWB NLOS Identification and Mitigation Methods Based on Gramian Angular Field and Hybrid Fusion Neural Network

Changke Zhu, Hongmei Zhao, and Yukang Su

***Abstract*—Ultrawideband (UWB) signals offer strong penetration capabilities and high transmission rates, playing an important role in precise indoor positioning. However, the complex and dynamic nature of indoor environments often leads to nonline-of-sight (NLOS) propagation and severe signal attenuation, resulting in significant ranging and positioning errors. Currently, convolutional neural network (CNN)-based approaches have made some progress in NLOS identification tasks. Nevertheless, traditional CNNs suffer from performance bottlenecks owing to their lack of temporal feature modeling capabilities and limited representational diversity. To address this issue, this article introduces the concept of feature fusion and proposes a multifeature fusion CNN model that incorporates Gramian angular field (GAF) encoding. By applying GAF encoding, it converts a 1-D channel impulse response (CIR) signal into a 2-D image, capturing the temporal correlations within the signals. During model training, different CNN architectures are trained using these GAF images, and the best-performing network is selected to construct a dual stream CNN that extracts both the time domain features of the original CIR signals and the frequency domain features of the GAF images. Meanwhile, physical layer parameters such as the maximum growth amplitude of the received CIR signals (maxGrowthCIR), the amplitudes of the first and second paths near the first path (FP_Ampl1 and FP_Ampl2), and UWB signal noise are incorporated to build a multidimensional auxiliary feature set. These auxiliary features enhance the ability to distinguish NLOS scenarios that are difficult to identify using CIR features alone, further reducing NLOS errors. In addition, a hybrid fusion method is used to capture correlations among channel features, mitigating the risk of data drowning caused by differences in input feature dimensions. Simulation results demonstrate that the proposed method achieves outstanding classification performance in both binary and multiclass NLOS tasks, with accuracy exceeding 95%. Compared with traditional algorithms, the proposed method reduces the mean absolute error (MAE) and root mean square error (RMSE) of distance estimation by 47.14% and 36.08%, respectively, achieving a coefficient of determination (R^2) of 0.9493. Furthermore, based on the recognition results, the proposed method can reduce the RMSE fraction and the MAE of the distance error by 43.78% and 41.67%, respectively, compared with the existing methods. In the actual indoor positioning experiments, this method improves positioning accuracy by more than 83%, fully verifying its engineering application potential in complex indoor localization scenarios.**

***Index Terms*—Convolutional neural network (CNN), Gramian angular field (GAF), hybrid fusion, indoor positioning, nonline of sight (LOS), ultrawideband (UWB).**

I. Introduction

INDOOR positioning technology serves as a crucial support for the realization of the Internet of Things (IoT) and smart buildings. It is widely applied in various fields such as smart cities, smart homes, healthcare, and security surveillance, and holds broad development prospects [1], [2], [3], [4]. Ultrawideband (UWB), as a high-precision indoor positioning technology, has become the preferred choice in many application scenarios owing to its centimeter-level positioning accuracy and low power consumption [5], [6]. However, the application of UWB in indoor environments still faces many challenges [7]. As shown in Fig. 1 in nonline-of-sight (NLOS) scenarios, obstacles such as walls cause significant increases in positioning errors [8]. The so-called NLOS error refers to the uncertainty introduced when UWB signals, affected by obstacles or structures, reach the receiver via indirect paths such as multipath reflections or to further improve the accuracy of UWB indoor positioning, researchers have proposed various methods to reduce NLOS errors [9].

Traditional positioning algorithms include time difference of arrival (TDOA), time of arrival (TOA), and angle of arrival (AOA), as well as filtering techniques such as Kalman filtering and particle filtering [10], [11], [12], [13], [14]. However, in complex and dynamic indoor environments, traditional methods generally suffer from high computational complexity, large computational load, and poor adaptability to environmental changes. In recent years, with the development of deep learning technologies, the NLOS error problem has been partially alleviated [15]. Deep learning methods represented by convolutional neural network (CNN) and recurrent neural network (RNN) have demonstrated strong feature learning capabilities under NLOS conditions and are expected to significantly improve positioning accuracy [16], [17], [18]. Deep learning models can learn and identify the complex differences between line of sight (LOS) and NLOS scenarios from massive amounts of training data, enabling more precise NLOS identification and effectively reducing ranging errors. Currently, a variety of

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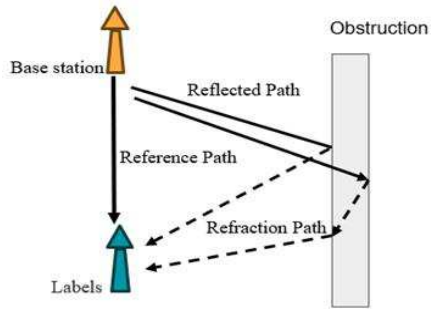


Fig. 1. UWB positioning system NLOS scenarios.

methods have been proposed for NLOS signal identification and mitigation. However, existing studies that employ CNN to extract channel impulse response (CIR) features still face issues such as loss of temporal characteristics and limited feature representation, making it difficult to accurately capture the complex nature of the original signals. Moreover, most existing methods heavily rely on prior knowledge and manual feature engineering of CIR signals, which limits their applicability in real-world scenarios. In addition, the majority of current research focuses only on simple binary classification between LOS and NLOS, lacking in-depth exploration of complex hybrid scenarios, thus struggling to effectively address increasingly complicated indoor environments. In most LOS/NLOS identification and scene recognition tasks, early fusion methods which combine multiple channel features directly at the input layer are commonly used. Owing to the large disparities in feature dimensions, direct fusion of CIR features with manually extracted channel features frequently results in data drowning, the dominant high-dimensional CIR features overshadow the impact of low-dimensional features, and important interfeature correlations may be overlooked [19]. To address these challenges, this article proposes an NLOS identification and error mitigation method based on GAF-encoded images, combining CNNs and a hybrid fusion strategy.

The main contributions of this article are summarized as follows.

1) Based on the characteristics of the collected experimental data and the functional requirements of the model, an innovative, improved CNN model was designed and compared with three traditional CNN models. Through rigorous analysis and evaluation, the optimal model was ultimately identified, demonstrating superior performance in handling multiclassification tasks related to UWB signal propagation channels.

2) A novel and efficient time series processing technique, Gramian angular field (GAF), was introduced to parse and convert CIR signals into RGB images, which serve as inputs for training and testing the improved CNN model.

3) Distinguishing itself from previous research, this study endeavored to tackle the challenge of NLOS identification in intricate hybrid environments, rather than confining the analysis to isolated single scenarios, making the research more conducive to addressing the complexities encountered in actual indoor positioning and sensing applications.

4) A hybrid feature fusion method was employed, which not only effectively captures the correlations among input features but also mitigates the risk of feature drowning caused by high-dimensional data. This approach enhances the identification performance for fine-grained NLOS scenarios with similar CIR characteristics, thereby further alleviating NLOS effects.

The remainder of this article is organized as follows. Section II briefly reviews previous research on NLOS identification and UWB systems. Section III presents the research background and preliminary explorations on UWB NLOS identification and error mitigation. Section IV introduces the LOS/NLOS identification method based on deep learning and hybrid fusion techniques. Section V provides a detailed description of the experimental design, results, and analysis. Section VI summarizes the main conclusions of this article and discusses directions for future research.

II. Related Work

LOS/NLOS identification and mitigation methods can be classified into two categories: those using machine learning and those that do not. The following will briefly introduce these two categories. Traditional methods analyze CIR signals using manually extracted statistical features methods that do not use machine learning. For example, Yang et al. [20] developed a high-precision indoor UWB positioning system based on inertial principles, using IMU data to estimate virtual sensor coordinates for compensating for NLOS-induced measurement errors. Wang et al. [21] combined inverse estimation with an improved unscented Kalman filter, integrating gyroscope and accelerometer data to identify and to compensate for NLOS occlusion errors. Sun et al. [22] proposed a robust positioning method by introducing a balance parameter and solving a weighted least squares optimization problem to jointly estimate source location and error terms. Yu et al. [23] presented an NLOS identification and mitigation approach independent of environmental prior knowledge, using a constrained Taylor series robust least squares technique to suppress residual errors. In addition, Che et al. [24] designed an NLOS detection algorithm based on Gaussian and generalized Gaussian distributions of signal features. Some researchers employed additional equipment like IMU sensors. Li et al. [25] integrated UWB with IMU, enhancing positioning accuracy in complex environments via maximum likelihood estimation and extended Kalman filtering. Silva and Hancke [26] distinguished LOS/NLOS using threshold values based on distance residual statistics. Other studies, such as [27] and [28], improved UWB positioning by constructing probability density function (pdf) models or multialgorithm optimization strategies. However, these traditional methods rely on manual feature extraction and static threshold settings. As a result, they struggle to adapt to dynamically complex environments and often exhibit low efficiency and poor robustness.

In recent years, machine learning technologies have offered new approaches for NLOS identification. Traditional methods, such as support vector machines (SVMs), aim to reduce manual feature engineering and improve accuracy. For example, Yang et al. [29] developed a two-step SVM classification

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method using dynamic threshold comparison and fuzzy confidence levels. Wei et al. [30] utilized neural networks to extract features directly from CIR and its time frequency representations. Olejniczak et al. [31] analyzed CIR parameter correlations with NLOS conditions, applying deep learning for classification. Semi-supervised and unsupervised SVM methods have also been used to address labeled data scarcity [32], [33]. While these methods enhance accuracy, they rely on handcrafted features, miss deep signal information, and require extensive preprocessing. To overcome these limitations, deep learning approaches using raw CIR signals for NLOS identification have become a focus. Lin et al. [34] introduced a CNN module into the autoencoder, utilizing CNN's strong ability to capture spatial features to accurately extract complex features such as multipath and Doppler in the ocean channel, thereby improving estimation accuracy. Liu et al. [35] integrated CNN and GRU modules with a squeeze-and-excitation mechanism to boost spatiotemporal feature extraction. Wang and Yang [36] evaluated SVM, multilayer perceptron (MLP), *K*-nearest neighbors (KNNs), and XGBoost in CIR classification tasks. Wang and Ahmad [37] applied feature selection and fuzzy evaluation before XGBoost classification. Studies [38], [39] used long short-term memory (LSTM) and CNN LSTM architectures to analyze temporal features. Zhang et al. [40] proposed a method combining empirical mode decomposition, wavelet denoising, and an improved CNN to reduce noise interference. However, LSTM modules often cause overfitting in small datasets. Pei et al. [41] adopted a fully convolutional network (FCN) with self-attention to improve accuracy, and Wang et al. [42] converted 1-D CIR time series to 2-D images via wavelet packet transform for CNN training. CNN, as an emerging technology, has been successful but often ignores measurement noise and temporal dependencies. Deng et al. [43] proposed a wavelet GAF CNN, combining wavelet transform and GAF encoding to suppress noise and extract trend features. Although deep learning achieves high accuracy, practical applications face challenges like the lack of standardized wavelet basis selection criteria, limiting generalizability. GAF, effective for encoding time series as images in mechanical diagnosis, remains understudied in UWB NLOS identification [44], [45], [46]. By preserving temporal dependencies and enriching signal representations, GAF provides strong data support for deep learning models, demonstrating significant potential.

In recent years, late feature fusion methods have demonstrated strong performance in multimodal recognition tasks. For example, their applications in high-speed railway scene recognition [47], [48] have significantly improved model performance. However, late fusion may miss potential feature correlations. To address this limitation, hybrid fusion methods have been proposed. These methods integrate the advantages of both early-stage and late-stage fusion, achieving promising results in tasks like waste classification [49], speaker recognition [50], and multimedia event detection [51], [52]. In LOS/NLOS recognition—where feature relationships are complex and semantically rich—hybrid fusion methods effectively capture intrinsic feature associations, minimize information loss, and enhance classification accuracy and robustness.

In summary, there are obvious limitations in the research on NLOS identification and mitigation. On the one hand, existing studies on NLOS identification and mitigation mainly focus on distinguishing LOS from NLOS scenarios but give less attention to detailed NLOS condition categorization. Therefore, this article selects four representative scenarios, covering the most common obstacle types and propagation characteristics in indoor environments, which can fully reflect the complex and diverse indoor communication environment and provide a reliable evaluation basis for the effectiveness of the method. On the other hand, Current methods often rely on prior knowledge, limiting the effective use of raw CIR signals for NLOS processing and reducing their practical applicability. To address these gaps, this article proposes a new NLOS identification and mitigation approach. The method uses GAF encoded images and an improved CNN with hybrid feature fusion. It accelerates model convergence by fully utilizing UWB CIR signals without requiring prior environmental knowledge. The approach integrates temporal and visual features, extracts UWB physical layer parameters to construct a multidimensional auxiliary feature set, and enhances model adaptability. Moreover, it accurately identifies multiple indoor NLOS scenarios and applies this information for mitigation, thereby improving localization performance in NLOS environments. The proposed method is consistent with the core objective of the current research, both taking UWB CIR signals as the core analysis object, aiming to achieve NLOS recognition and error mitigation by mining signal features. This is in line with the mainstream direction of "NLOS processing based on CIR signals" in the field. In addition, the method adopts the concept of automatically extracting features with a deep learning model. In line with the research trend in recent years of "processing CIR timing signals with CNN and LSTM" which aims to reduce reliance on manual feature engineering.

Traditional nonmachine learning methods rely on manual extraction of static parameters or setting thresholds, are limited by prior knowledge, and are difficult to adapt to dynamic scenarios. The proposed method converts 1-D CIR into 2-D images through GAF, combines the multifeature hybrid fusion (MFHF) model to automatically mine temporal and visual features, eliminates the need for manual feature design, effectively reduces the reliance on prior knowledge, and significantly improves processing efficiency and generalization performance. Traditional machine learning requires manual selection of feature vectors and cannot fully utilize the deep temporal information of CIR. The proposed method takes the original CIR and GAF images as input and extracts features end-to-end through CNN and deep neural network (DNN). The physical layer parameters, such as the maxGrowthCIR of the signal, the FP_Ampl1, the FP_Ampl2, and the noise of the UWB signal, are introduced as auxiliary features to avoid the limitations of manual feature selection and achieve higher feature utilization. Compared with the existing deep learning methods, in terms of feature transformation approaches, the current studies either directly handle 1-D CIR, which has the drawback of easily losing features, or use wavelet transform (CWT) to convert 2-D images but require manual selection of

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wavelet bases (with poor adaptability). The proposed method applies GAF to NLOS recognition, generating 2-D images through polar coordinate transformation and Gramian matrix, fully preserving the CIR temporal correlation without the need for manual parameter adjustment, and thereby addressing the issues of feature loss in 1-D signals and the limited adaptability of CWT. In terms of model structure, most existing deep learning models are single-input architectures, with single feature extraction and prone to overfitting. The MFHF model extracts CIR temporal features, GAF image visual features, and auxiliary features, respectively, through three parallel modules, and then fuses multimodal features through a hybrid fusion method to enhance the comprehensiveness of features and the stability of the model, effectively alleviating overfitting. In terms of scene obstacle handling, most existing methods are limited to the binary classification of LOS and NLOS, without further subdividing NLOS scenes caused by different obstacles. The proposed method subdivides NLOS into scenarios such as filing cabinets, concrete columns, and iron doors, making error correction more targeted and solving the problem of insufficient adaptability of traditional methods to complex environments.

In summary, the method proposed in this article is based on the existing deep learning processing of CIR signals. Through GAF transformation, multimodal fusion, and multi-NLOS scene perception, it breaks through the feature extraction limitations and scene adaptability bottlenecks of traditional methods, providing an innovative reference for UWB high-precision positioning in complex indoor environments.

III. Preliminary Analysis

A. Extraction of UWB Parameters for Different Obstacle Environments

Within the UWB transmission architecture, omnidirectional antennas are used to exchange signals between two devices. The signal propagation process is susceptible to environmental interference, leading to distortions due to obstruction or reflection.

CIR characterizes signal transmission aberrations because it reflects the channel's response to short pulse signals, allowing for an accurate description of communication behavior. In a UWB-based positioning system architecture, the data transmission process between the anchor node and tag node can be shown in the following equation:

$$s(t) = \sqrt{M}\sum_{n=0}^{N-1} p\left(t - nT_p\right). \tag{1}$$

In this expression, M represents the value of the amplitude of a single pulse, p is the pulse waveform, N denotes the total number of pulses used to construct the information frame, and T_p is the length of the sampling interval. The UWB channel pulse model can also be modified to be

$$h(t) = \sum_{i=1}^{I} \varphi_i \delta(t - \beta_i). \tag{2}$$

Owing to the presence of delays and fading in the different paths, reflected in the coefficients φ_i and β_i, the signal received by the UWB sensor is essentially the cumulative result of multiple individual pulses which can be expressed as

$$r(t) = \sum_{i=1}^{I} \varphi_i s(t - \beta_i) + k(t). \tag{3}$$

In the above expression, $k(t)$ is the additive Gaussian white noise.

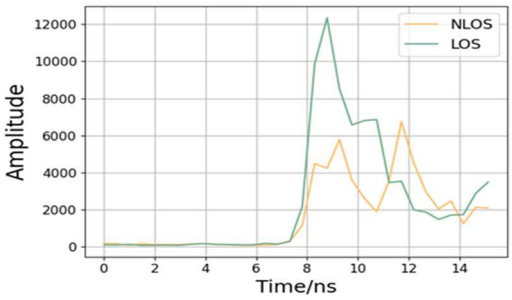


Fig. 2. UWB CIR signals at LOS and NLOS conditions.

The first path (FP) data in the CIR of UWB signals is important, as it typically corresponds to the direct path of signal arrival, usually characterized by the shortest propagation time and, in ideal conditions, the strongest signal. By extracting FP information, the impact of multipath effects can be effectively minimized, thereby improving positioning accuracy and system robustness. The directness, anti-interference capability, and stability of the FP make it a crucial reference point for CIR analysis, especially in complex environments. In practical applications, FP data has been proven to significantly enhance the performance of positioning systems, providing reliable support for real-time localization based on UWB technology. In this study, the CIR signal is derived from the received waveform $r(t)$, specifically, eight data points before and after the FP are extracted. From a total of 16 000 CIR signal samples, 100 samples are randomly selected and averaged. Experimental comparative analysis shows that the essential characteristics of the CIR signal can be represented using the data around the FP. This sampling strategy ensures that the extracted data sufficiently captures CIR features while avoiding the inefficiency caused by excessive data volume. Moreover, the use of random sampling and averaging helps reduce noise and improves model robustness. A standard UWB CIR curve is illustrated in Fig. 2, where a clear distinction between CIR curves under LOS and NLOS conditions can be observed, demonstrating the potential to distinguish LOS from NLOS scenarios.

The operational stability of UWB-based measurement systems fundamentally depends on reliable signal transmission between anchor and tag nodes. Accurate distance measurement requires spatially referenced anchor nodes with known coordinates. Among various ranging techniques, the time of flight (TOF) method is especially effective, as it directly calculates signal propagation time. Assuming the measurement is only influenced by NLOS factors, the distance between the anchor node and the tag node measured by the TOF method can be

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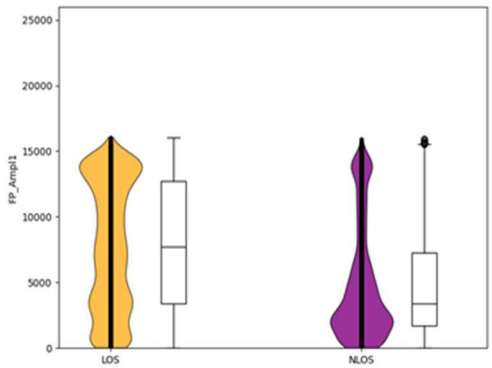


Fig. 3. Violin box plot of maximum increase in CIR.

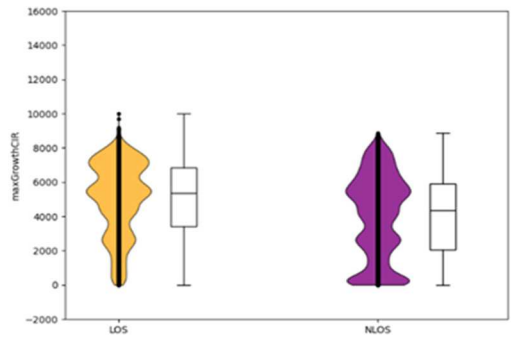


Fig. 4. Violin box plot of the FP amplitude near the FP.

presented by the following expression:

$$d = 0.5c\left[(t_d - t_a) - (t_c - t_b) + 2\tau_i\right]. \tag{4}$$

In the above equation, c denotes the propagation rate of the signal, t_a is the timestamp corresponding to when the anchor point first sends a CIR signal to the tag, t_b is the timestamp when the tag successfully receives the signal sent by the anchor point, t_c is the timestamp marking of when the tag sends a response signal back to the anchor point, and t_d is the timestamp when the anchor point acquires the response signal sent by the tag, and τ_i represents the signal transmission time delay caused by the NLOS condition.

The ranging information of UWB is obtained using an FP edge detection algorithm that dynamically detects FP information on the CIR. Before the signal reaches the receiver, the CIR image exhibits a noisy pattern, similar to that seen in the first half of Fig. 2. Once the signal arrives, the system identifies the FP as the first peak exceeding a predefined threshold. In a LOS scenario, the FP corresponds to the strongest peak in the CIR. However, in an NLOS scenario, the path mistakenly identified as the FP often lags behind the actual direct path, resulting in signal delay and energy attenuation. In addition, due to dynamic changes in the communication environment and multipath effects introduced by NLOS, diagnostic parameters of the UWB system's internal registers may also experience fluctuations.

The analysis of collected data reveals distinct differences in CIR parameters between LOS and NLOS environments. Key parameters, including maxGrowthCIR, the amplitudes of the

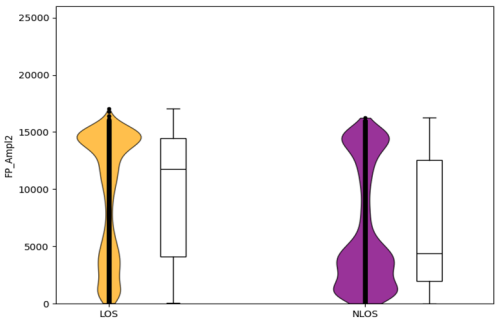


Fig. 5. Violin box plot of the second path amplitude near the FP.

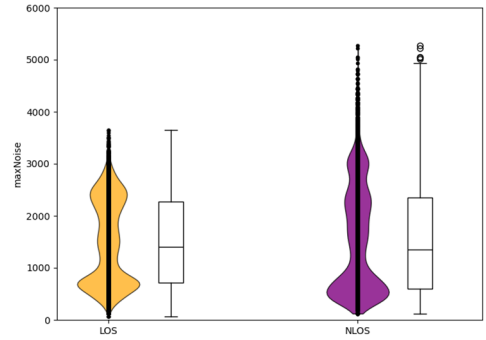


Fig. 6. Violin box plot of the maximum noise of the UWB signal.

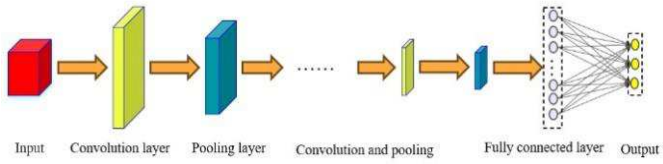


Fig. 7. CNN architecture diagram.

first and second paths in the vicinity of the primary signal path, and UWB signal noise levels, show notable variations, as illustrated in Figs. 3–6. Detecting these variations enables the recognition of environmental changes by analyzing CIR data. Since CIR signals are time series data, their scenario-dependent characteristics allow machine learning models to effectively differentiate between LOS and NLOS propagation conditions.

B. Convolutional Neural Network

As shown in Fig. 7, a typical CNN architecture comprises five core components: input layer, convolutional layer, pooling layer, fully connected layer, and output layer. The network depth and layer-specific hyperparameters require customization based on specific application requirements. Feature map dimensions progressively decrease through sequential processing stages, ultimately generating predictions via the fully connected layer. The input layer converts raw data into numerical arrays through preprocessing. In the convolutional layer—the architecture's central component—multiple feature maps containing neurons utilize convolutional kernels to

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TABLE I
SELF-BUILT CNN_2-D STRUCTURE

Layer	Hyper parameter	Activation shape	Layer parameter
INPUT		(128,128,3)	0
CONV1	f = 3, s = 1, p = same, n = 64	(128,128,64)	1792
POOL1	f = 3, s = 2, p = same	(64,64,64)	0
CONV2	f = 3, s = 1, p = same, n=16	(64,64,16)	9232
BN1	$\beta_0 = 0$, $\gamma_0 = 1$, $\varepsilon = 1e-3$	(64,64,16)	0
POOL2	f = 3, s = 2, p = same	(32,32,16)	0
FC3		(128,1)	2,097,280
FC4		(128,1)	16,512
SM5		(2,1)	257

TABLE II
SELF-BUILT CNN_1-D STRUCTURE

Layer	Hyper parameter	Activation shape	Layer parameter
INPUT		(128,3)	0
CONV1	f = 3, s = 2, p = same, n = 32	(64,32)	320
BN1	$\gamma = 1$, $\beta = 0$, ε = 1e-3	(64,32)	0
POOL1	f =3, s=2, p=same	(32,32)	0
FC2		(1024,1)	2,098,176
FC3		(256,1)	262,400
SM4		(2,1)	513

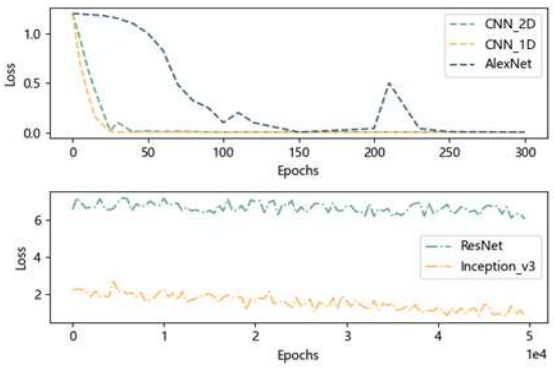


Fig. 8. Training loss of five CNN models.

extract localized patterns from input data. Activation functions then transform these neuron inputs into outputs. Following feature extraction, the pooling layer reduces feature map resolution through operations like max pooling, average pooling, or stochastic pooling. This dimensionality reduction achieves three objectives: computational load minimization, dominant feature retention, and overfitting prevention during training. The fully connected layer integrates features extracted across spatial locations by establishing complete interlayer neuron connectivity. This global synthesis mechanism enhances network performance by combining localized patterns. Finally, the output layer produces task-specific results for classification or regression applications.

Compared with other machine learning algorithms, CNN has distinct advantages in feature learning. Especially when dealing with image inputs, it achieves lower computational complexity, offers a more consistent and noise-resistant feature extraction process, and effectively reduces the risk of model overfitting.

IV. PROPOSED METHODOLOGY

This study first conducted a comparative evaluation of two custom CNN architectures under the same experimental conditions. Benchmark comparisons were conducted with the established classical models AlexNet [53], ResNet [54], and Inception_v3 [55], to verify the superiority of the proposed architecture and select the optimal configuration for the subsequent stage. A novel GAF transformation method was used to convert CIR signals into visual representations, enabling deep exploration of embedded spatial–temporal patterns. The optimized network extracted both GAF-derived image features and raw temporal characteristics from CIR data. These heterogeneous features were integrated with DNN outputs through a multistage fusion strategy. Finally, we detailed the architectural framework and operational workflow of the MFHF system, specifying component interactions and feature integration mechanisms.

A. Performance Analysis and Selection of UWB Signal Propagation Channel Classifier

We developed two custom CNN architectures, named CNN_1-D and CNN_2-D. They differ structurally in the number of layers, learning rate, layer-specific training parameters, and the output dimension of the activation function, as shown in Tables I and II. By comparing the results with those of benchmark models, we determined the optimal architecture for the classification of UWB signal propagation channels. The GAF technique transformed CIR signals into 128 × 128 RGB energy maps, which served as standardized network inputs. These visual representations preserved both temporal and spectral characteristics of the raw CIR signals, enabling effective feature learning through convolutional operations.

During model training, 70% of the data was used as the training set for the five CNN models, and the remainder was used for testing and validation. The learning rate controls parameter updates during training, and the batch size determines the number of samples processed per iteration. Since

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both parameters affect CNN performance, identical learning rates and batch sizes were assigned to all models. The loss function measures the alignment between predicted and actual classification labels—lower loss values indicate smaller prediction errors, while stable values indicate model convergence. To track changes in loss for the five CNNs, the error curves were visualized using a Python visualization tool, as shown in Fig. 8, training Loss of Five CNN Models. The two custom CNN models achieved a rapid reduction in loss to near zero and exhibited faster convergence speeds. In contrast, AlexNet required more training cycles, while ResNet and Inception_v3 exhibited relatively long convergence periods, with ResNet stabilizing at a loss of 7.2 and Inception_v3 at 1.4.

This study considers two architectures, CNN_1-D and CNN_2-D. The core purpose is to identify the CNN architecture most suitable for the multiclassification task of UWB signal propagation channels by comparing models with different structural complexities. The specific reasons are as follows: UWB CIR is a 1-D signal in the time domain, containing temporal characteristics such as the arrival time and amplitude of multipath components. CNN_1-D is selected for its streamlined structure and compatibility with time-series data. To adapt to the characteristics of GAF-converted input data, the UWB CIR is transformed into a 128 × 128 pixel RGB color coefficient image after being processed by the GAF. This type of image contains the energy distribution characteristics of UWB signals in LOS and NLOS channels. Traditional CNN models, mainly designed for natural images, often suffer from structural redundancy or limited targeted feature extraction. However, self-built CNN_1-D and CNN_2-D can customize their structures according to the specific time series and the dimensions and features of the images. This customization helps avoid low training efficiency or overfitting caused by excessive complexity. A comparison of different structural complexities reveals significant differences in structural complexity between CNN_1-D and CNN_2-D: CNN_2-D consists of 2 convolutional layers, 2 pooling layers, 1 batch normalization (BN) layer, and 2 fully connected layers, while CNN_1-D includes 1 convolutional layer, 1 pooling layer, 1 BN layer, and 2 fully connected layers. Compared with deeper models, CNN_1-D is less prone to overfitting due to the redundancy of time-series data. At the same time, the computing efficiency is higher. By comparing the training convergence speeds of the two, the relationship between "structural complexity and task adaptability" can be clarified, providing a basis for selecting the optimal model. Balancing recognition accuracy and computational efficiency, traditional models such as ResNet and Inception_v3 have complex structures with multilayer convolution and residual connections, making it difficult to meet the efficiency requirements. However, CNN_1-D and CNN_2-D have a streamlined structure and short training time, making them more suitable for practical applications.

The parameter settings of both models are based on the characteristics of the input data, feature extraction requirements, and experimental verification optimization, as follows: for the 1-D temporal characteristics of CIR, CNN_1-D takes an input dimension of 128 × 3. The convolution kernel f = 3 of CNN_1-D can capture the local temporal correlation, the amplitude variation of three adjacent sampling points, and adapt to the continuous time distribution of multipath components. The step size s = 2 is used to compress the temporal dimension and reduce redundant sampling points. At the same time, the key time-window characteristics are retained. Max pooling extracts the amplitude peaks within the local timing window, the strongest response of the multipath components, and enhances the peak difference between the LOS FP and the NLOS multipath. BN stabilizes scale fluctuations of time series features and the amplitude differences of CIR in different scenarios, accelerating model convergence. The dimension is adjusted according to the temporal feature dimension output by the convolutional layer to avoid overfitting caused by too many parameters, while ensuring that it can be mapped to the output of three types of channels. The size of the color coefficient image processed by CNN_2-D for GAF is 128 × 128, and the color mode is RGB. This parameter ensures the complete retention of the multipath energy distribution characteristics of CIR, providing effective input for subsequent convolution operations. The convolution kernel size of 3 × 3 is the optimal size for extracting local features of the image, which can capture the color intensity changes of adjacent pixels. The color intensity changes reflect the differences in energy distribution. The CNN_2-D stride is set to 1 with "same" padding to retain the feature-map size of the feature map to avoid the loss of early energy distribution features, which is particularly crucial for the wide distribution features of NLOS channels. The number of convolution kernels is set to 64 in the first layer are used in the first layer to extract rich color gradients of the underlying features, and 16 convolution kernels are used in the second layer to balance feature diversity and computational load. Max pooling can extract local salient features and the color coefficients corresponding to the energy peaks, which is more suitable for distinguishing the energy concentration of LOS from the multipeak differences of NLOS. The 3 × 3 pooling window and step size 2 compress the feature map while following the standard implementation of BN, training convergence is accelerated by normalizing the mean and variance, reducing model instability caused by feature scale fluctuations, which is particularly crucial for small-sample training. The fully connected layer maps the features extracted by convolution to the category space, preparing for subsequent classification and feature fusion.

B. Multipart Figures

Time series reconstruction techniques are extensively applied across various fields, particularly in machine learning. This study introduces an innovative approach that leverages GAF technology to transform CIR signals into RGB images. The method preserves essential CIR signal characteristics while using CNNs to analyze signals efficiently.

The GAF coding process consists of three specific steps. First, the original CIR signal $X = [x_0, x_1, x_2, \ldots, x_n]$ should be normalized to the range between 0 and 1. This normalization

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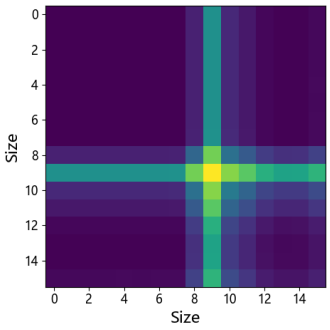


Fig. 9. LOS processed by GAF.

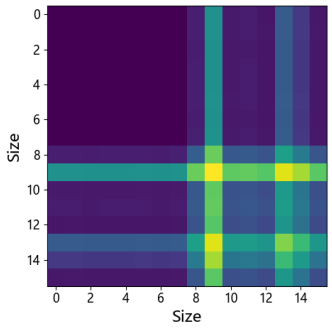


Fig. 10. NLOS processed by GAF.

process is expressed as follows:

$$\widetilde{x_i} = \frac{x_i - \min(X)}{\max(X) - \min(X)}, \quad i \in [0,n]. \tag{5}$$

Then, the normalized signal $\widetilde{X} = [\widetilde{x_0}, \widetilde{x_1}, \widetilde{x_2}, \ldots, \widetilde{x_n}]$. When converting the Cartesian coordinates to polar coordinates, the conversion process is expressed in the following form:

$$\begin{cases} \emptyset_i = \arccos(\widetilde{x_i}), \quad 0 \le \widetilde{x_i} \le 1 \\ \mathrm{r}_i = \dfrac{t_i}{N} \end{cases} \tag{6}$$

In the above transformation equation, $\emptyset_i$ is the angular component, r_i represents the radius, while t_i corresponds to the timestamp of x_i, and N represents a constant value. Together, $\emptyset_i$ and r_i determine the specific range and position in the polar coordinate system. Through such a setting, the normalized signal $\widetilde{X}$, which is originally in Cartesian coordinates, can be presented in polar coordinates in the form of a specific angle and radius, thus providing a new perspective and data basis for the subsequent in-depth analysis and processing of CIR signals.

After the polar coordinate transformation, the obtained parameters are fed into the Gramian angular summation fields (GASFs) function. The result is a Gram matrix, where its elements represent the image pixels. This process converts the CIR signal into a 2-D matrix. This process can be modeled as follows:

$$G_{ij} = \cos\left(\emptyset_i + \emptyset_j\right), \quad i, j \in [0,n] \tag{7}$$

$$\mathrm{GASF} = \begin{pmatrix} G_{11} & \cdots & G_{1n} \\ \vdots & \ddots & \vdots \\ G_{n1} & \cdots & G_{nn} \end{pmatrix}. \tag{8}$$

In the GASF method, G_{ij} represents the elements of the Gram matrix, where n is the length of the time series. Equations (7) and (8) are expressed using the triangular transformation as follows:

$$\mathrm{GASF} = \widetilde{X}' \cdot \widetilde{X} + \sqrt{I - \widetilde{X}^2} \cdot \sqrt{I - \widetilde{X}^2}. \tag{9}$$

Figs. 9 and 10 show the GAF-transformed CIR signals for LOS and NLOS, respectively. The conversion process fully retains the original signal data while creating visual representations. Notably, lighter-colored elements emphasize distinguishing features of NLOS conditions, while the remaining elements represent background noise. Visual analysis reveals distinct pattern differences between LOS and NLOS signals after GAF encoding. These visual discriminators enable effective signal classification and processing, offering valuable insights into UWB signal propagation under different environmental conditions.

Based on the forgoing discussions of CNN and GAF, this article proposes an innovative combination of the two methods. The method proceeds as follows: in the data input stage, after generating images of different signal propagation channels through GAF, they are not directly used as input to the CNN model. Instead, data preprocessing is first performed. A portion of the image dataset is allocated for CNN training, while the remaining samples serve as validation and test sets. Data labeling is performed while ensuring the integrity of original features, thus significantly reducing manual labeling costs.

The process proceeds to the convolutional layer, which essentially performs a specialized matrix operation between the input image and the convolutional kernel. Specifically, each weight of the convolution kernel f is multiplied with the pixel value at the corresponding position in the input image I, and the resulting products are summed. This process can be mathematically expressed as

$$G(i,j) = (I * f)\left[i,j\right] = \sum_m \sum_n f(m,n)\, I(i-m, j-n). \tag{10}$$

Here, m and n represent row and column indices in the convolution kernel matrix, while i and j correspond to row and column indices of the convolutional layer's output feature map. The convolutional layer involves three critical hyperparameters: kernel sizes, strides, and paddings. Convolution kernels usually use odd-dimensional square matrices. The number of kernels depends on the number of input and output channels. In this case, the input consists of three channels. Stride defines the sliding distance of the kernel during operation. Larger strides typically cause greater feature loss during image processing. To mitigate feature degradation in multilayer convolution, padding adds dummy pixels to the edges of the feature maps. Two padding types exist: "Same" padding maintains input and output dimensional consistency, whereas "Valid" padding applies no edge extension.

The pooling layer is applied immediately after the convolutional layer to reduce the spatial dimensions of feature maps through down-sampling. This operation lowers computational complexity while preserving salient features. Two main types

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of pooling are commonly used: max pooling and average pooling. Max pooling selects the highest value from each local region, whereas average pooling computes the mean value within the region. In this study, max pooling is employed due to its effectiveness in capturing prominent features. Similar to convolutional layers, max pooling requires the specification of three hyperparameters: pooling window size, stride, and padding. The window size determines the spatial extent of the pooling operations. The stride controls the step size of the window during traversal, impacting both feature resolution and computational efficiency. Padding is applied to preserve the spatial dimensions and prevent information loss at the boundaries. Proper parameter tuning enhances pooling efficiency and overall model performance.

To improve the training speed and generalization ability of CNN models, a regularization technique known as BN is introduced. In comparison to other normalization approaches, such as local response normalization, BN normalizes the inputs to each layer to maintain a stable mean and variance. This mitigates issues such as vanishing or exploding gradients, thereby significantly accelerating training and enhancing model generalization. Let a mini-batch Ω of size m be defined as = $\{a_1, \ldots, a_m\}$, where $a_1, \ldots, a_m$ denote the input activations. The essence of BN can be described as follows:

$$\begin{cases} \dfrac{1}{m}\displaystyle\sum_{i=1}^{m} a_i \to \mu\Omega \\ \dfrac{1}{m}\displaystyle\sum_{i=1}^{m} (a_i \to \mu\Omega)^2 \to \sigma\Omega^2 \\ \dfrac{(a_i \to \mu\Omega)}{\sqrt{\sigma\Omega^2 + \varepsilon}} \to \hat{a}_i \\ \mathrm{BN}_{\gamma,\beta}(a_i) \equiv \gamma\hat{a}_i + \beta \to y_i. \end{cases} \tag{11}$$

Let $y_i\,(i = 1, 2, \ldots, m)$ denotes the output activation, where γ and β are the learnable scale and shift parameters, respectively. These parameters are updated during training. A small constant ε is added to the denominator to ensure numerical stability and to avoid division by zero. The BN operation normalizes the inputs and then scales and shifts the normalized values using γ and β. This process stabilizes training, accelerates convergence, and enhances the model's generalization capability across different datasets. As a result, the model is better equipped to handle unseen data with improved predictive performance. Further details about BN can be found in [56].

The final components in a CNN architecture are the fully connected layer and the Softmax layer. The fully connected layer transforms the extracted features into a K-dimensional output vector, whose elements can take any real value within the range $(-\infty, +\infty)$. Each neuron in this layer is densely connected to all neurons in the preceding layer. Feature integration and transformation are performed through matrix multiplication followed by bias addition. This process can be mathematically expressed as follows:

$$G = W^T x + b. \tag{12}$$

Let W^T denote the transpose of the weight matrix of the fully connected layer, x the input feature vector from the previous layer, and b the bias vector. The output G is a K-dimensional vector resulting from the linear transformation. This mapping aggregates the local features extracted by the preceding convolutional and pooling layers into a compact and discriminative global representation, which serves as the input for the final classification stage.

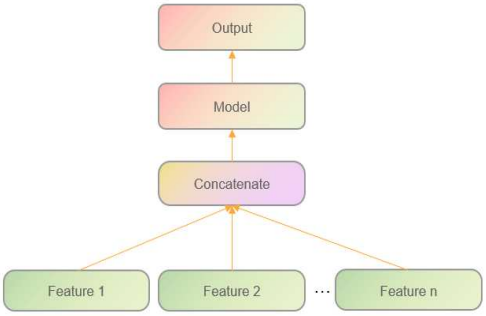


Fig. 11. Early fusion framework.

The Softmax layer is used to map the K real numbers in the range $(-\infty, +\infty)$ to real numbers in the range (0, 1). The Softmax output determines the predicted class of the input sample. Accordingly, the probability that a GAF-transformed test image X belongs to class l can be expressed as

$$p(X, l) = \mathrm{softmax}(gl) = \frac{e^{gl}}{\sum_L e^{gl}} \tag{13}$$

$$\text{subject to} \sum_L p(X, l) = 1 \tag{14}$$

where gl denotes the lth element of the output vector G, and L is the total number of target classes. After obtaining class probabilities from the Softmax layer, the predicted class of the test image X is determined by selecting the class with the highest probability. Hence, the final predicted label for X is given by

$$P(X) = \mathrm{index}(\max\{y(X, l)\}) \begin{cases} 0, & X \in \mathrm{NLOS} \\ 1, & X \in \mathrm{LOS}. \end{cases} \tag{15}$$

Here, index(u) denotes the position of class u in the output vector. This index information enables the identification of the specific category assigned to the test image.

Based on the classification results, the type of signal propagation channel corresponding to the test image can be determined. Through the integration of CNN and GAF, and after a sequence of data preprocessing, feature extraction, and classification, images corresponding to different signal propagation channels can be accurately classified. This lays a foundation for further research and practical deployment of UWB signals in diverse propagation environments.

C. Mixed Fusion Processing

Using a single feature representation often results in suboptimal classification performance. To improve the recognition accuracy of LOS/NLOS conditions, feature fusion strategies are widely adopted. Existing feature fusion approaches are generally categorized into three types: early fusion, late fusion,

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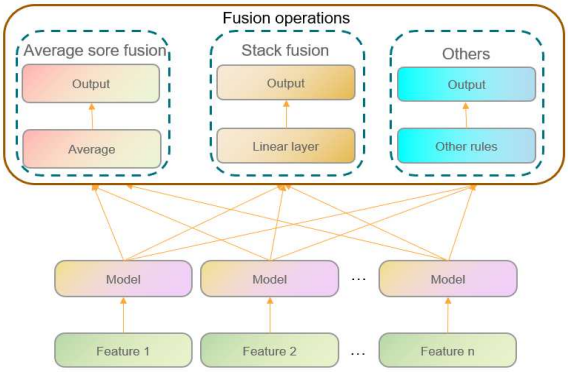


Fig. 12. Late fusion framework.

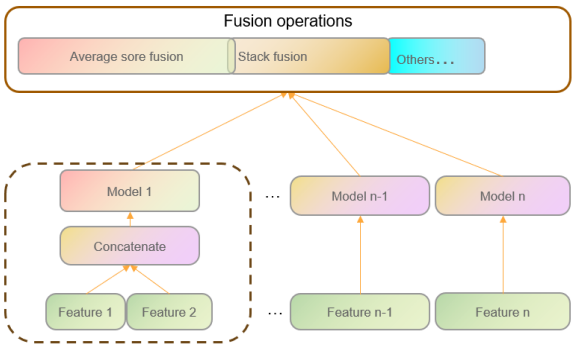


Fig. 13. Hybrid fusion framework.

and hybrid fusion [57]. As illustrated in Fig. 11, early fusion concatenates correlated features into a high-dimensional vector, which is then fed into a single classifier. Although easy to implement, this approach may suffer from dimensional dominance, particularly when combining high-dimensional and low-dimensional features. Such interference can obscure the contribution of low-dimensional features, thereby degrading overall classification performance.

Late fusion, also known as decision-level fusion, involves training separate models on different feature streams, followed by the fusion of their outputs, as shown in Fig. 12. Common fusion strategies include arithmetic averaging and weighted stacking [58]. In this study, two approaches are employed: score averaging, which calculates the mean of prediction scores from multiple models to produce a final decision, and stack fusion, which transforms prediction outputs via linear layers prior to applying Softmax classification. Compared with early fusion, late fusion benefits from model-specific learning of feature representations, thereby reducing the risk of dimensional dominance. However, this method requires training multiple models independently, resulting in higher computational cost and more frequent parameter updates. The overall computational complexity corresponds to the aggregate of the individual model complexities, leading to increased resource consumption and extended training time.

Hybrid fusion integrates early and late fusion strategies to exploit their respective strengths. As Fig. 13 illustrates, this architecture contains two interconnected components.

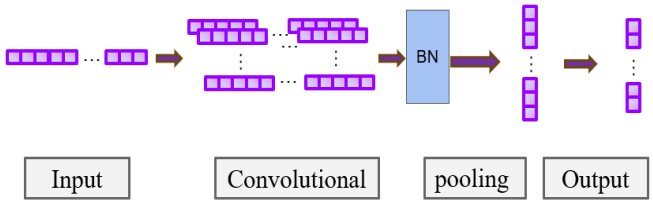


Fig. 14. CIR time domain feature extraction framework.

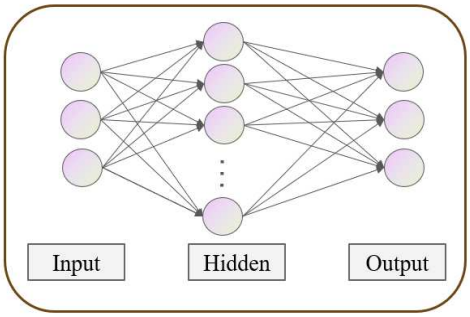


Fig. 15. Auxiliary feature extraction framework.

The early fusion component selects and concatenates feature subsets as input to a shared base model. Concurrently, the late fusion component processes remaining features through independent models. The outputs from these two modules are subsequently integrated at the decision level. This design enables flexible adaptation to feature characteristics, preserving feature dependencies while mitigating the risk of dimensional dominance. Compared with pure fusion approaches, hybrid fusion reduces the number of required models through optimized feature allocation, thereby lowering computational complexity. Practically, this balanced approach addresses early fusion's interference risks and late fusion's resource demands, thereby improving the accuracy of LOS/NLOS classification.

D. Feature Extraction

Building upon the prior coverage of RGB feature extraction, this section focuses on temporal feature extraction. CNNs demonstrate strong performance in time-series classification, especially when handling long sequences or when data availability is limited. This study employs CNN_1-D for temporal feature extraction from CIR signals. As Fig. 14 illustrates, the CNN_1-D architecture contains three core components: First, a convolutional layer processes temporal patterns. Second, BN and max pooling layers accelerate convergence and mitigate vanishing gradients. Finally, flattening and dense layers convert features into compact vectors that preserve multimodal temporal information, enabling effective CIR signal representation and classification.

The auxiliary feature extraction module utilizes a DNN to automatically capture complex nonlinear relationships within the data, as illustrated in Fig. 15. This three-layer architecture (input, hidden, and output layers) processes key CIR signal

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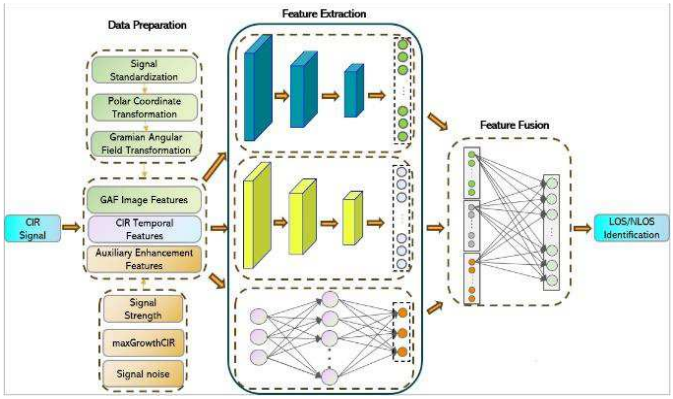


Fig. 16. Framework of the FMHF CNN model.

features, including maximum signal rise, energy in proximity to the FP, and noise distribution patterns. By integrating these engineered features with learned representations, the module enhances scene classification accuracy. It also helps mitigate the impact of NLOS propagation. In particular, UWB-derived signal characteristics contribute to accurate characterization and attenuation of NLOS effects in wireless channel analysis.

The feature vectors obtained from the three extraction modules are rescaled and concatenated to form a unified feature representation. This fused vector is then fed into a hidden layer with a Softmax activation function to perform classification. During model training, the cross-entropy loss function is used to evaluate classification performance and monitor potential overfitting. In addition, the learning rate is automatically adjusted based on the loss function's output, effectively mitigating overfitting risks.

The overall workflow of the proposed method is illustrated in Fig. 16. The proposed MFHF model is a deep learning framework that integrates multiple sources of features to enable NLOS recognition and error mitigation. Its overall structure comprises three modules, which correspond to three processing stages. The specific processes and technical details of each stage are as follows: The first stage is signal preprocessing, in which timing features are visualized using the GAF, with the goal of transforming the 1-D CIR time-domain signal into a 2-D image to enhance the representation of temporal correlations. First, the CIR signal is normalized and scaled to the range $[-1, 1]$, as defined in (5). The foundation is laid for the subsequent polar coordinate transformation. Next, a polar coordinate transformation is performed, mapping the standardized signal to the formulation in (6), where $\emptyset_i$ preserves the amplitude characteristics, and r_i preserves the temporal information of the original signal. Subsequently, GAF image generation is performed, generating a 2-D image through the GAF. GAF reflects the correlation of signal angles and sums at different times through (7), which can comprehensively capture the temporal correlation of signals. At the same time, the temporal features of the CIR are extracted, as well as auxiliary enhancement features such as signal strength, maxGrowthCIR, and signal noise, to comprehensively obtain signal-related information. Then, the feature extraction stage begins, where a parallel neural network structure is adopted to conduct in-depth mining of GAF image features, CIR time features, and auxiliary enhancement features, respectively. The visual feature extraction module (CNN_2-D) is based on a CNN and is used to process 2-D images converted from GAF to extract visual features. It consists of two convolutional layers, two pooling layers, and a BN layer, gradually mining spatial features in the image and accelerating training convergence. This reduces model instability caused by fluctuations in feature scales. The output is processed by the flattening layer and the dense layer to ensure matching with the output of other modules. The time feature extraction module (CNN_1-D) is built on a CNN and is specifically designed to handle 1-D CIR signals for extracting temporal features. It consists of a convolutional layer that uses 32 convolutional kernels to capture the local temporal correlations of CIR signals through convolution operations. After the convolutional layer, a max pooling layer is connected to reduce the feature dimension. It retains key information, while BN layers to alleviate vanishing gradients and accelerate convergence; The auxiliary feature extraction module, DNN, is designed to learn the mapping relationship between manually extracted low-dimensional channel features and LOS/NLOS conditions. Features such as signal strength, maxGrowthCIR, and signal noise are concatenated into a vector as the input of the DNN. The hidden layer is a fully connected layer with 64 neurons, and the parametric PReLU activation function is employed as the activation function can effectively alleviate the problem of neuron "death." The weight initialization adopts the He initialization method to avoid the disappearance of the neuron activity of the PReLU activation function. The output features, after processing, participate in the fusion together with the features of the CNN_1-D and CNN_2-D modules. Different modules correspond to distinct processing branches for different feature types. Through multilayer network operations, key information that can effectively represent scene differences is extracted from various features. Finally, in the feature fusion stage, a hybrid fusion strategy is adopted, combining the advantages of early fusion and late fusion. The initial early fusion of CIR images and temporal features during the extraction process enables the model to capture the underlying correlations between features earlier. Subsequently, the extracted auxiliary features are further fused in the late stage to fully integrate the high-level semantic information of features from different dimensions. This not only utilizes the ability of early fusion to capture fine-grained associations but also achieves efficient integration of abstract features through late fusion. The fused features are then processed through fully connected layers and additional network structures, and the final LOS or NLOS recognition results are output, providing a basis for applications such as scenario-specific UWB positioning error correction. The entire framework enhances the accuracy and robustness of LOS/NLOS recognition through the mixed fusion and deep extraction of multisource features.

V. Experimental Design and Analysis of Results

This section first introduces the dataset, followed by the selection of standard evaluation metrics to assess the classification and error prediction performance of the NLOS

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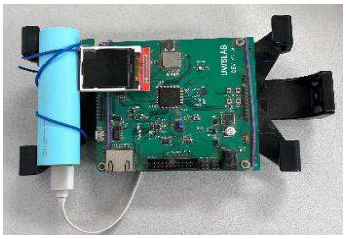


Fig. 17. UWB anchor.

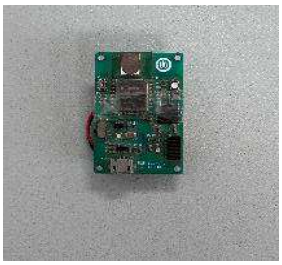


Fig. 18. UWB tag.

system. We purposely select the representative methods in the field of traditional machine learning, as well as deep learning, CNN, and SVM, to demonstrate the robustness of the proposed framework. Experiments also examine the effect of multifeature fusion on error mitigation. Finally, practical validation in complex indoor localization scenarios confirms the method's effectiveness and applicability.

A. Experimental Setup and Data Collection

The experiment used the DWM1000 UWB transceiver module from Decawave, which complies with the IEEE 802.15.4a standard. The hardware configurations of the anchor and tag nodes are shown in Figs. 17 and 18. Key UWB signal parameters were set as follows: channel 2, a 128-byte preamble, 64-MHz pulse repetition frequency, 8-byte packet length, and a data rate of 6.8 Mbps. As the core component of the positioning system, the DWM1000 chip features a high degree of integration, low cost, and low power consumption. This transceiver chip supports both transmission and reception and enables TOF-based ranging, including TOA and TDOA techniques. The UWB signal receiver was connected to a computer via a USB interface to collect communication log files. Data processing and algorithm implementation were performed on a Windows 11 64-bit workstation equipped with an Intel Core i5-8250U CPU (1.60 GHz base frequency), 12-GB RAM, a GTX1050 GPU, and a 128-GB SSD. The system environment included MATLAB 2016b and Python 3.12.0 for software support.

The model may remain effective under certain conditions even if the user employs hardware specifications different from those used in the experiment. Specifically, the experiment utilized a UWB device based on the DW1000, with hardware parameters such as packet channel 2; 128-byte preamble code; 64-MHz pulse repetition frequency; 8-byte data packet length; and 6.8-Mbps data rate. All measurements were conducted following the IEEE 802.15.4a UWB channel model.



Fig. 19. Underground parking lot scenario.



Fig. 20. Laboratory building corridor scenario.



Fig. 21. Laboratory scenario.

The core of the model is NLOS identification and mitigation through CIR signal processing, and its effectiveness relies on the inherent channel propagation characteristics captured in the CIR signals rather than on hardware-specific parameters. As long as the UWB hardware can generate CIR signals conforming to the general UWB channel model, the model can extract the LOS/NLOS propagation characteristics—such as the prominent first-path peak in LOS scenarios and significant multipath interference in NLOS scenarios—enabling effective recognition and mitigation. By focusing on the common features of CIR signals, the model demonstrates adaptability within the range of hardware conforming to the general UWB standard.

Three experimental environments were used for evaluation: a laboratory at Zhengzhou University of Light Industry, a corridor, and a parking lot. Device placement followed established criteria as described in [59] and [60], with deployment

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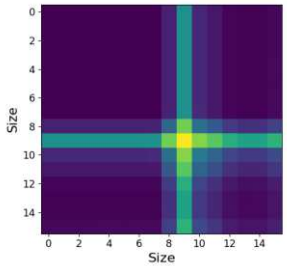


Fig. 22. Unobstructed GAF encoded image.

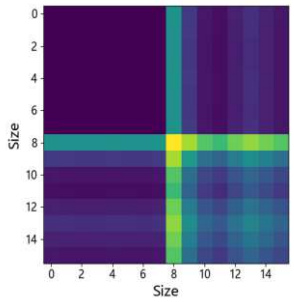


Fig. 23. GAF encoded image with door obstruction.

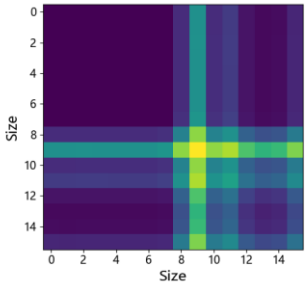


Fig. 24. GAF encoded image with pillar obstruction.

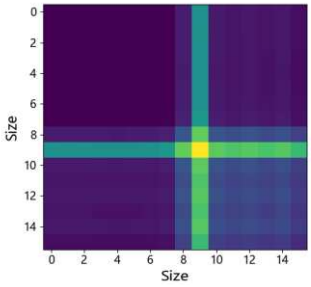


Fig. 25. GAF encoded image with filing cabinet obstruction.



Fig. 26. Unobstructed condition.

configurations illustrated in Figs. 19–21, with anchor points placed at known coordinates and the same height. A total of 8000-UWB signal waveforms were collected across diverse propagation channels to ensure data diversity. A systematic labeling process categorized all received signals. The dataset was randomly divided into 70% as the training subset and 30% as the test subset.

The CNN model was trained to learn signal features and construct a propagation channel classifier. First, for data processing, to reduce the computational complexity, sixteen points before and after the FP of the CIR were captured and encoded into a 128 $\times$ 128 $\times$ 3 image using GAF. This process involves three steps. The first step is to normalize the original CIR signal to between 0 and 1. The second step is to convert the normalized signal from Cartesian coordinates to polar coordinates. The third step is to input the parameters obtained after converting to polar coordinates into the GAF function. The result of the function forms a Gramian matrix, whose elements can be regarded as pixels that make up the image. This matrix is input into the CNN_2-D network. After processing by the convolutional layer, the data becomes 128 $\times$ 128 $\times$ 64, and then processed by the pooling layer, which reduces data volume and enhances computational efficiency. Then, perform a convolution to lower the number of channels from 64 to 16. Introduce BN to normalize each channel. Next, carry out a one-step pooling operation. Finally, expand it into 1-D feature data through a fully connected layer. Similarly, the UWB CIR data is processed by CNN_1-D to obtain the same 1-D CIR time series feature data. After processing the UWB data through box-plot analysis, the signal features that help distinguish NLOS from LOS are extracted by using a DNN with one hidden layer and 128 neurons, for example, the maximum increase in CIR, the FP amplitude near the FP, and the UWB signal. The three types of extracted features are fused together through the means of mixed fusion to form a new vector. Finally, the new feature vectors are, respectively, input into a fully connected output layer with a Softmax activation function for classification. The cross-entropy loss function was used to guide optimization, and the learning rate was adaptively adjusted based on the loss, thereby effectively alleviating overfitting.

During the data collection process, tags were moved within an indoor environment containing typical obstacles in order to capture CIR signals exhibiting a range of propagation characteristics. Simultaneously, a laser range finder was used to measure the actual distance between each tag and anchor, which served as the ground truth. The CIR signal's propagation time was then used to estimate the corresponding distance. By comparing these estimated values with the true distances obtained from the laser range finder, the measurement error could be quantified. To mitigate overfitting caused by location-specific data distributions, all collected samples were randomly shuffled. In total, 16 000 samples were randomly partitioned into training (70%) and test (30%) datasets. Figs. 22–25

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Fig. 27. Door obstruction condition.



Fig. 28. Pillar obstruction condition.



Fig. 29. Filing cabinet obstruction condition.

display GASF-encoded images corresponding to the LOS scene and three different NLOS scenarios, respectively. Figs. 26–29 present the actual photographs of these scenes, providing visual context for the subsequent analysis.

B. Evaluation of Performance Indicators

In the evaluation of NLOS recognition models for UWB signals, four evaluation metrics that are widely used in the research field of classification prediction, namely, accuracy, recall, precision, and $F1$ score, are adopted in this study. For clarity and conciseness, these metrics are denoted as $P_{\mathrm{Ac}}, P_{\mathrm{Re}}, P_{\mathrm{Pr}}$, and P_{F1}, respectively, throughout this article. All metrics are defined based on true positives (TPs), true negatives (TNs), false positives (FP), and false negatives (FNs). Specifically, TP indicates that the signal is NLOS and predicted as NLOS; TN denotes the signal is LOS and predicted as LOS; FP means the signal is LOS but predicted as NLOS; and FN means the signal is NLOS but predicted as LOS. The definitions and corresponding formulas for each evaluation metric are provided below.

Accuracy measures the overall classification performance of the model and provides a general indication of its ability to correctly classify UWB signals as either LOS or NLOS. It is calculated as follows:

$$P_{\mathrm{Ac}} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{16}$$

Recall, also known as sensitivity, measures the classifier's ability to accurately identify signals that are actually in NLOS. It is calculated using the following formula:

$$P_{\mathrm{Re}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{17}$$

The accuracy represents the classifier's ability to correctly predict NLOS signals in all samples. It is calculated using the following formula:

$$P_{\mathrm{Pr}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{18}$$

The $F1$ score is used to evaluate the overall performance of the classifier. The closer the recall and precision are, the higher the $F1$ score will be. The $F1$ score is the harmonic mean of recall and precision, balancing the tradeoff between the two. It is calculated by the formula

$$P_{F1} = \frac{2 \times P_{\mathrm{Re}} \times P_{\mathrm{Pr}}}{P_{\mathrm{Re}} + P_{\mathrm{Pr}}}. \tag{19}$$

This article evaluates the NLOS error regression model for UWB signals using three common metrics: mean absolute error (MAE), root mean square error (RMSE), and R^2, to assess the performance of different models. Their calculation formulas are as follows:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left| y_i - y'_i \right|. \tag{20}$$

MAE represents the average of the absolute differences between the true and predicted values of each sample. It reflects the average magnitude of the residuals predicted by the regression model. Let y_i be the true value of the sample, y'_i the predicted value, and N the number of samples. By averaging the absolute differences between the true and predicted values of each sample, MAE provides an intuitive measure of the model's average deviation from the true values. A smaller MAE indicates that the model's predictions are closer to the true values, reflecting higher prediction accuracy

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(y_i - y'_i \right)^2}. \tag{21}$$

RMSE is calculated by first squaring the prediction error for each sample, averaging the squared errors, and then taking the square root of the result. This process increases the penalty for larger errors, making it more sensitive to differences between the predicted and actual values. A smaller RMSE indicates a better fit of the regression model to the data and more accurate predictions

$$R^2 = \frac{\frac{1}{N} \sum_{i=1}^{N} \left(y_i - y'_i \right)^2}{\frac{1}{N} \sum_{i=1}^{N} (y_i - \bar{y}_i)^2} \tag{22}$$

R^2 quantifies the fit of the regression model by measuring the proportion of variance in the dependent variable explained by

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TABLE III
BINARY CLASSIFICATION RESULTS OF DIFFERENT MODELS

Method	Accuracy	Precision	Recall	F1
DNN	75.1	70.3	78.4	74.1
SVM	77.4	75.9	79.4	77.6
CNN	89.4	91.3	86.9	89.3
MFHF	95.82	95.88	95.84	95.87

the independent variables. The value of R^2 ranges from 0 to 1. An R^2 of 0 indicates that the model fails to explain the variation in the dependent variable, and there is no correlation between predicted and actual values, suggesting a poor fit. In contrast, an R^2 value close to 1 implies that the model effectively explains the variance, and its predictions align closely with actual outcomes, indicating a strong model fit.

To comprehensively evaluate the MFHF model's binary classification performance, it is compared with three benchmark methods: DNN, SVM, and CNN. As presented in Table III, the performance of each method in LOS/NLOS classification is assessed using four standard metrics. The MFHF model consistently exceeds 90% across all metrics, outperforming conventional models in distinguishing between LOS and NLOS scenarios. Further analysis reveals that CNN achieves an accuracy of 89.4%, which is higher than 75.1% of DNN and 77.4% of SVM. The relatively lower performance of SVM can be attributed to its limited capacity in capturing high-dimensional CIR signal features. Despite this, SVM still demonstrates certain advantages, particularly in low-dimensional feature spaces. The study leverages a dataset of approximately 16 000 CIR samples, which contributes to effective overfitting mitigation and improved classification robustness.

The selected control schemes, including traditional machine learning methods such as SVM and deep learning methods such as DNN and CNN, are chosen based on the current research status and technical characteristics in the field of NLOS identification and mitigation. The rationale for their selection can be explained as follows: the control schemes mentioned in this article are all mainstream methods in the field of NLOS recognition and mitigation, covering technical routes from traditional machine learning to deep learning. The specific selection basis is as follows: traditional machine learning methods were commonly used schemes for early NLOS recognition. Previous studies have applied traditional machine learning methods, such as SVM to NLOS recognition, which rely heavily on manual feature selection. Choosing SVM as a control directly demonstrates the advantages of MFHF in automatic feature extraction, highlighting the superiority of deep learning over traditional methods. The single-input deep learning method, CNN, is effective in handling the local features of time-series signals. CNN has been widely used to extract temporal features from the original CIR signals and is a classic model for processing 1-D time-series data in NLOS recognition. The image feature processing model CNN_2-D has also been introduced in the article. Some studies convert CIR into 2-D images through GAF transformation and then process them with deep learning. CNN is a classic model for image feature extraction. By choosing CNN as a control, the effects of "single visual feature extraction" and MFHF's "temporal + visual feature fusion" can be specifically compared, highlighting the advantages of multimodal fusion. Selecting these two types of models allows comparison of the performance differences between single feature extraction and MFHF's multifeature parallel fusion.

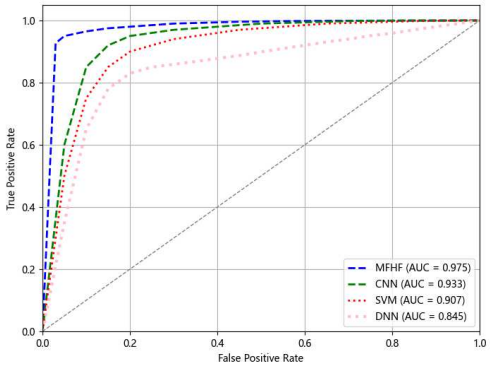


Fig. 30. ROC curves of different models.

The control schemes cover traditional machine learning, image processing models, and single-input deep learning. This design comprehensively reflects the innovations of MFHF in feature utilization (manual versus automatic), model structure (single-input versus multi-input), and modal fusion (single versus multimodal), thus forming a gradient contrast. The selected methods are all directly related to the core innovation points of MFHF. Compared with SVM, automatic feature extraction is superior to manual feature selection. Compared with CNN, the multi-input parallel structure is superior to the single-input architecture, and temporal and visual feature fusion outperforms single visual feature extraction. These methods have also been widely cited in NLOS recognition research. In the work of Sang et al. [61], different machine learning tools were applied; SVM, RF, and MLPs were used to construct LOS/NLOS classifiers. Stahlke et al. [62] pointed out that SVM cannot fully utilize the features of CIR, resulting in lower recognition accuracy compared with CNN. Choosing them as controls conforms to research conventions in the field and makes the results directly comparable.

The phrase "not choosing specific existing mature solutions" has certain limitations, but this should be understood in light of the research objectives: the core innovation of MFHF lies in the multimodal feature fusion architecture and GAF-based time-to-image conversion, while existing mature solutions are mostly based on a single feature, such as CIR timing or RSS, which differs significantly from the design concept of MFHF. Choosing the basic ML/DL structures as controls allows us to eliminate confounding factors and directly highlight the structural innovation of MFHF and GAF transformation. If a dedicated optimization scheme for UWB NLOS were added, the comprehensiveness of the comparison could be further enhanced. However, the existing controls already effectively support the validity of the core innovation points.

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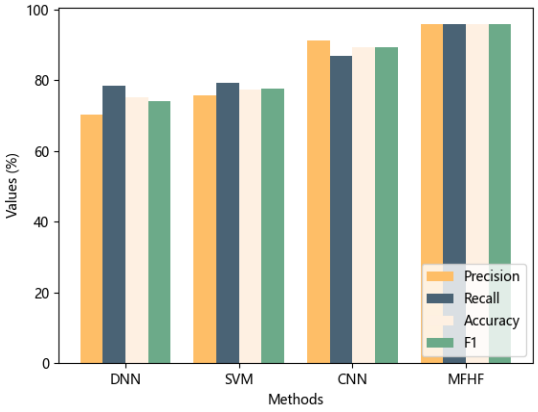


Fig. 31. Classification performance in complex scenarios.

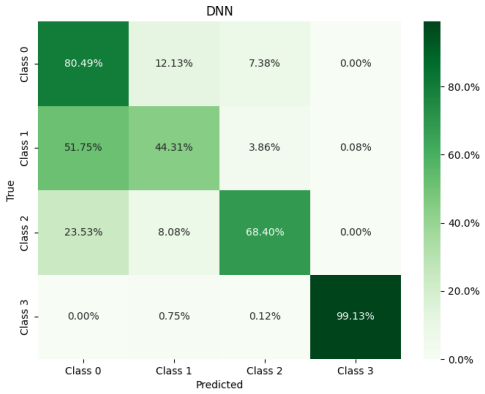


Fig. 32. Confusion matrix of CNN classification.

In addition, to further assess the performance of the MFHF model, Fig. 30 presents the receiver operating characteristic (ROC) curve. The MFHF model achieves a significantly higher area under the curve (AUC) compared to conventional methods, indicating superior capability in distinguishing between LOS and NLOS scenarios. A higher AUC value reflects a more effective classifier, thereby highlighting the model's robust performance in binary classification tasks. In summary, the integration of temporal and visual features significantly improves the reliability and precision of LOS/NLOS classification. While the MFHF model exhibits robust performance under constrained experimental conditions, further validation is required to assess its generalization capability in more complex and dynamic real-world environments.

To further evaluate the performance of the MFHF model in complex scene classification tasks, a multiscene classification test was conducted. The indoor environment is characterized by complex spatial structures. Typical building structures include concrete walls, columns, doors, and glass [63]. Columns that play a load-bearing role in building structures are generally made of steel and concrete and are relatively large in size. The resulting NLOS propagation is highly complex, affecting a wide range of UWB signals and exhibiting unstable error patterns [64]. Zhang et al. [65] demonstrated the great assistance of positioning in the unique and complex indoor environment through in-depth analysis of structural relationships and the integration of multisource feature information. Four scenarios were selected in this study, covering one LOS scenario and three NLOS scenarios. In terms of the core types of obstacles in the indoor environment covered by the scene, the three selected NLOS obstacles are concrete columns, filing cabinets, and metal doors. They are the most common obstacles in indoor environments and cover typical interference sources of different materials, structures, and electromagnetic characteristics. Concrete columns, which represent solid obstacles in the main structure of a building, can significantly block UWB signals and generate strong reflections. The materials of filing cabinets, including wood and glass, are common indoor partitions and lighting structures, which reflect the attenuation characteristics of nonmetallic solid and semi-transparent materials, respectively. Metal doors and other metallic obstacles, due to their high conductivity, induce strong signal reflections and absorption, simulating electromagnetic interference in scenarios such as warehouses and laboratories. These obstacles cover the main interference factors of signal propagation in indoor environments (reflection, refraction, scattering, and attenuation), and are representative of the propagation characteristics in most indoor environments. From the verification and analysis of the distinguishability of scene features, the experimental results show that the CIR signals corresponding to different obstacles have unique characteristic contours, as illustrated in Figs. 22–25, which can meet the requirements of NLOS/LOS recognition. Meanwhile, the average NLOS recognition accuracy for the three obstacle types reached 98%, indicating that the signal characteristics of these scenarios are highly distinguishable and effectively capture propagation differences in complex indoor environments. In conclusion, the four selected scenarios cover the most common types of obstacles and their propagation characteristics in indoor environments, which can fully represent the complex and diverse indoor propagation environment and provide a reliable evaluation basis for the effectiveness of the method. Fig. 31 illustrates the precision, recall, accuracy, and $F1$ score achieved by three traditional models and the proposed MFHF model. The results indicate that conventional approaches suffer significant performance degradation in multiclass classification tasks, particularly DNN and SVM, which exhibit the most notable declines across all evaluation metrics. This performance drop reflects limitations in their feature extraction capabilities, making it difficult to accurately distinguish between different NLOS scenes. In contrast, the MFHF model maintains strong classification performance and substantially outperforms the baselines. Specifically, compared with CNN, MFHF achieves improvements of 6.42% in accuracy, 4.58% in precision, 8.94% in recall, and 6.57% in $F1$ score. These results confirm the effectiveness and robustness of the proposed model in complex indoor classification scenarios.

Confusion matrices provide detailed insights into model classification performance. As shown in Fig. 32, the CNN model shows moderate accuracy but struggles to distinguish

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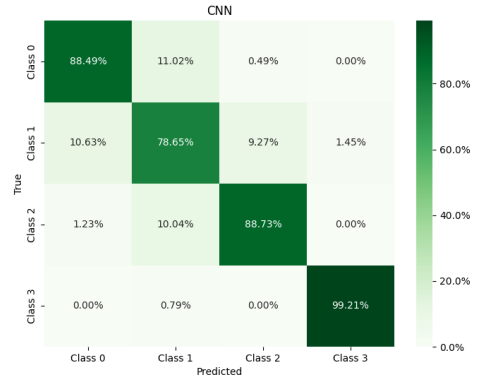


Fig. 33. Confusion matrix of DNN classification.

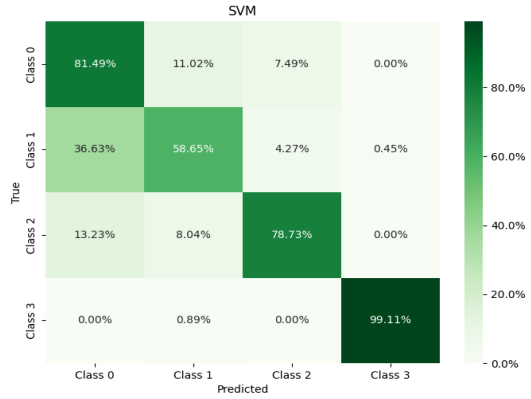


Fig. 34. Confusion matrix of SVM classification.

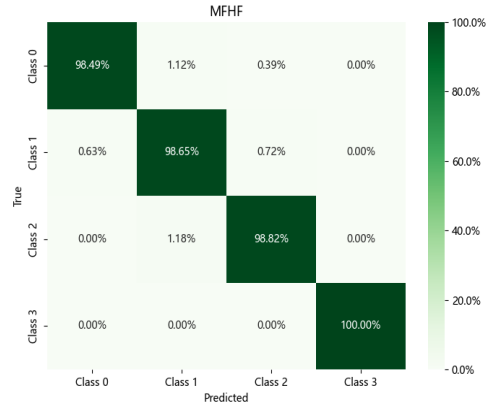


Fig. 35. Confusion matrix of MFHF classification.

similar scenes, especially Classes 0 and 1. Figs. 33 and 34 indicate that DNN and SVM suffer from more severe misclassification, with some classes nearly indistinguishable. However, as shown in Fig. 35, the proposed MFHF model demonstrates superior performance in multiclass classification, achieving perfect classification accuracy for Class 3. Notably, MFHF significantly improves recognition accuracy for classes 0, 1, and 2 as well, thereby enhancing the overall classification reliability. These results highlight MFHF's effectiveness in complex scene classification.

TABLE IV
MODEL CLASSIFICATION PERFORMANCE COMPARISON

Method	MAE	RMSE	R^2
DNN	0.2433	0.3329	0.467
SVM	0.2387	0.3297	0.529
CNN	0.1983	0.2806	0.786
MFHF	0.1286	0.2128	0.9493

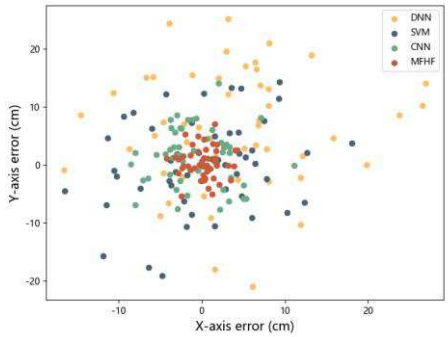


Fig. 36. Error distribution scatter plot.

To comprehensively evaluate the effectiveness of the proposed method in mitigating NLOS errors and to investigate the benefits of segmenting NLOS scenarios, subsequent experiments compare the performance of the MFHF method with several commonly used distance error mitigation approaches. This comparison aims to highlight the advantages of the proposed segmentation strategy in enhancing ranging accuracy under complex propagation conditions.

Table IV presents a comprehensive comparison of various NLOS error regression models. The proposed MFHF model demonstrates superior performance across key evaluation metrics, outperforming baseline models built solely with CNN, DNN, or SVM. Specifically, MFHF reduces the MAE to 0.12 m and the RMSE to 0.2128 m. Compared to DNN, SVM, and CNN, MFHF achieves a reduction in MAE of 47.14%, 46.13%, and 35.15%, respectively. In terms of RMSE, it yields corresponding reductions of 36.08%, 35.46%, and 24.16%. Moreover, MFHF attains an R^2 value of 0.9493, significantly higher than 0.467 of the comparison model DNN, 0.786 of CNN, and 0.529 of SVM. These results confirm that the proposed feature fusion strategy significantly enhances the regression accuracy of the MFHF model in NLOS error mitigation tasks.

Fig. 36 shows the error distributions for the four methods. Fig. 37 shows the probability distribution curves of distance errors for four methods. DNN and SVM show a wider error distribution, indicating lower stability compared to CNN and the proposed MFHF algorithm. MFHF significantly surpasses other algorithms in terms of localization accuracy and stability. Localization errors for CNN mostly fall below 20 cm, with a significant portion under 15 cm. In contrast, MFHF's localization errors predominantly remain below 7 cm, with the

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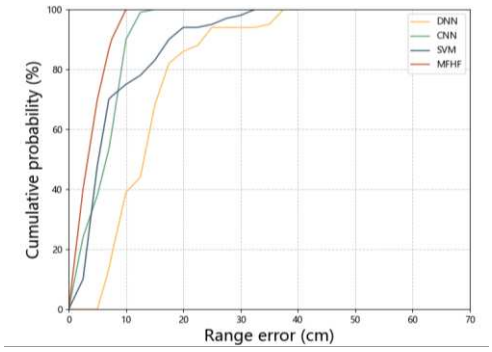


Fig. 37. Distance error cdf of different methods.

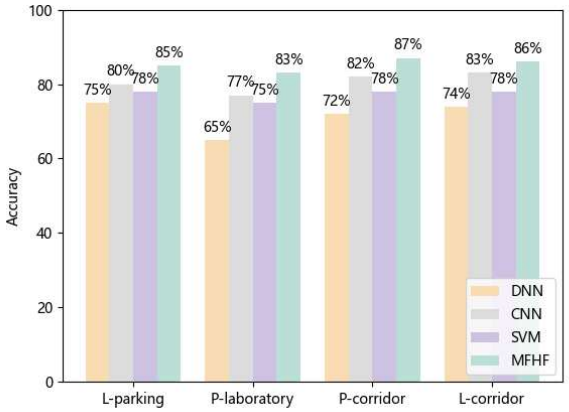


Fig. 38. Comparison of recognition accuracy in different scenarios under heterogeneous training.

majority being below 5 cm. The results for MFHF are tightly clustered around the origin (0, 0), demonstrating the highest localization accuracy among the evaluated methods.

C. Analysis of Experimental Results of NLOS Occlusion Recognition

The 16 000 CIR samples collected in this study (including both LOS and NLOS scenarios) were obtained from three indoor environments at Zhengzhou University of Light Industry—laboratories, corridors, and parking lots. The dataset was randomly divided into 70% for training and 30% for testing. To avoid overfitting caused by position-specific data, samples from different locations were randomly selected and mixed for training, ensuring that the test set samples did not participate in training. In this way, the test samples shared the same distribution as the training samples while remaining independent. This random partitioning and mixing process effectively ensured the independence of the training and test data and mitigated the risk of overfitting to specific environmental features. To accurately evaluate the classification performance of the UWB channel, experiments were conducted under two conditions: homologous and heterologous environments. A heterologous environment involves training and test data from different sites, while a homologous environment uses data from the same site. For instance, when evaluating the MFHF model trained on laboratory data, the laboratory serves as the training environment, while the parking lot is used as the test environment. They are denoted by their initial letters. If the training environment is a laboratory, we use the letter *L*; if it is a parking lot, we use the letter *P*. In the experiments, the laboratory and parking lot alternated as the measuring and nonmeasuring environments. Due to the relatively simple corridor environment in this article, it is not used as a training environment. This approach enables a comprehensive evaluation of the MFHF model's ability to classify UWB channels in diverse environmental scenarios. It also provides a precise assessment of its performance stability and reliability.

Data were collected from the laboratory, parking lot, and corridor to train the MFHF model and the other three models. The corridor, being relatively empty, is treated as a LOS case in this study and compared with the two more complex scenarios. To comprehensively evaluate model performance, two key aspects are analyzed: recognition accuracy and computation time. Recognition performance is measured in three dimensions: accuracy, precision, and recall. The accuracy rate, representing overall classification performance, is calculated as the ratio of correct recognitions to total attempts. Fig. 38 compares the accuracy rates of each model across different scenarios. MFHF achieves over 93% accuracy in all three scenarios, outperforming DNN, CNN, and SVM by average increases of 14.33%, 5.33%, and 8%, respectively. When comparing performance across locations, the parking lot scenario yields better results than the laboratory. This is mainly because the laboratory environment contains more obstacles, with obstacle density approximately twice that of the parking lot, as shown in Fig. 21. This leads to a reduction in environmental noise, which in turn lowers the signal-to-noise ratio. Consequently, this reduces the classification performance of the model. Notably, CNN accuracy decreases by approximately 5% in the laboratory, whereas MFHF accuracy decreases by about 4%. More severe environmental noise intensifies the impact on feature extraction. In real-world environments, classification performance under different channel conditions follows a similar pattern. In addition, training and testing times were recorded to evaluate model efficiency: training time refers to the duration spent on training, and testing time refers to the duration for processing CIR data.

In summary, in the multiscenario classification test, NLOS conditions were subdivided into common indoor obstacles, including filing cabinets, doors, and columns, and the model underwent multiscenario classification testing. The results are shown in Figs. 32–35, indicate that the MFHF method outperforms other baseline models in overall accuracy and effectively distinguishes scenarios with similar CIR characteristics, demonstrating strong adaptability to various NLOS subscenarios. The MFHF model extracts UWB signal features as auxiliary inputs for challenging scenarios and employs segmented NLOS scene data for error mitigation tests. It achieves superior error mitigation compared to models without auxiliary features or segmented data, indicating robust performance across different NLOS scenarios. Cross-experiments were conducted under both homogeneous and heterogeneous

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TABLE V
EFFICIENCY EVALUATION OF DIFFERENT MODELS

Methods	Training Time(s)	Test time(s)
DNN	5.98	0.000416
SVM	3.58	0.000826
CNN	15.73	0.0000322
MFHF	264.78	0.000145

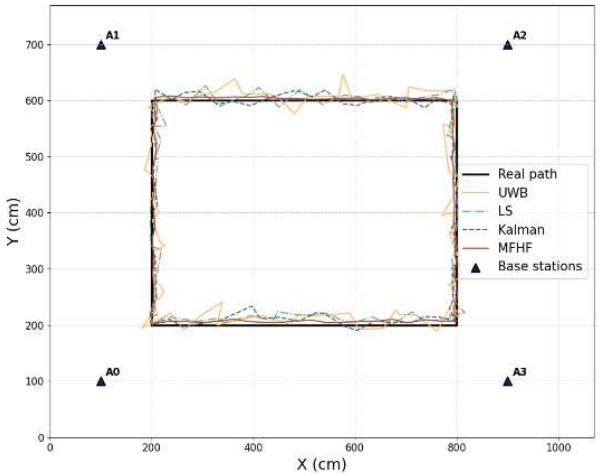


Fig. 39. Trajectory observed in each method of the underground parking scene.

TABLE VI
RMSE OF POSITIONING ERRORS FOR EACH METHOD IN THREE SCENARIOS

Method	Underground parking lot [cm]	Laboratory [cm]	Corridor [cm]	Average [cm]
UWB	21.86	28.64	13.99	21.497
LS	16.85	17.77	11.49	15.370
Kalman	13.27	14.53	10.71	12.837
MFHF	7.56	8.66	5.43	7.217

environments. As shown in Fig. 38, the results demonstrate that the model achieves higher accuracy across different sites compared with other algorithms.

The proposed MFHF method adopts a more complex three-layer architecture than conventional models, resulting in higher training complexity. As shown in Table V, CNN has a training time of 15.73 s, while MFHF requires 264.78 s, a significant difference. However, since training occurs prior to deployment, the impact on real-time operation is minimal. Posttraining, MFHF maintains low testing complexity, with inference times comparable to SVM, CNN, and DNN. Despite its training overhead, MFHF significantly enhances UWB localization accuracy and stability by more effectively processing signal data.

D. Localization Performance

This section evaluates the positioning performance of the designed system using the cdf, RMSE, and MAE of positioning errors across three trajectory types. Figs. 39–44 illustrate

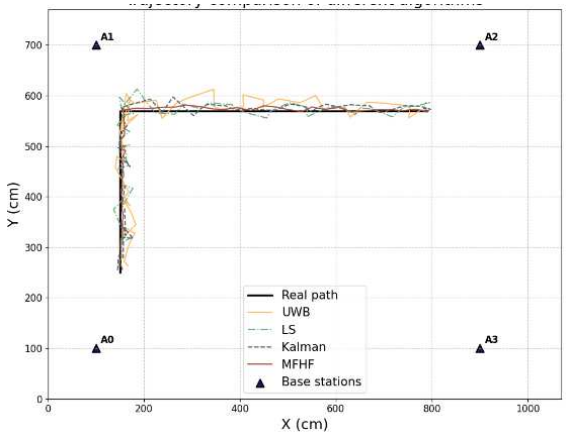


Fig. 40. Trajectories observed in each method in the laboratory scenario.

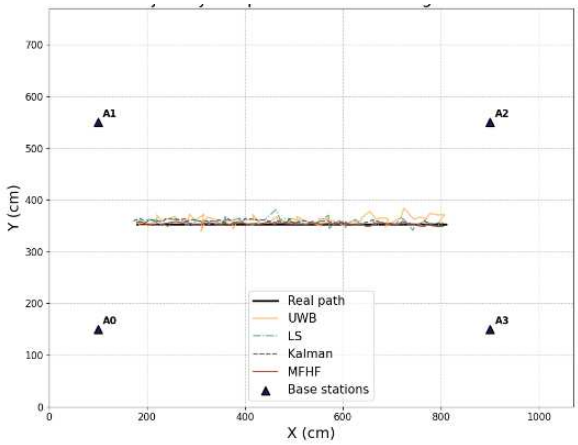


Fig. 41. Trajectory observed in each method of the corridor scene.

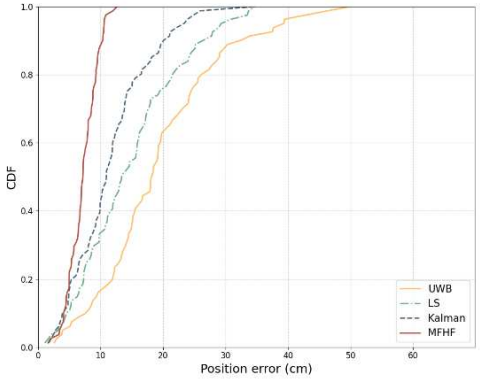


Fig. 42. CDF of positioning error measured for each method in underground parking lot scenarios.

the three positioning trajectories for each method and their corresponding CDFs. In addition, Tables VI and VII, respectively, show the RMSE and MAE of positioning errors measured by each method under the same conditions. In the experimental scenarios, each scene includes four fixed anchors and one moving tag. The true position of each anchor is measured by a laser rangefinder. The mark moves along the reference path. The UWB method and the LS method were compared with the Kalman method. The results indicate that the RMSE of the original UWB positioning is 21.497 cm, while those of LS, Kalman, and MFHF are 15.370, 12.837, and

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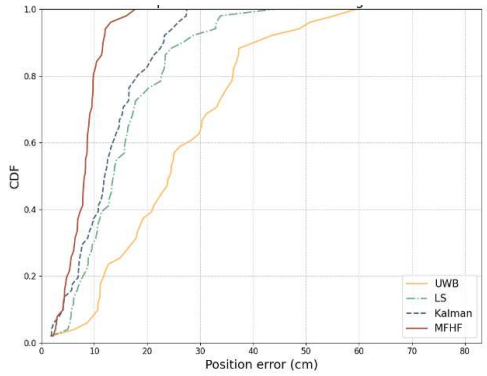


Fig. 43. CDF of positioning error measured for each method in laboratory scenarios.

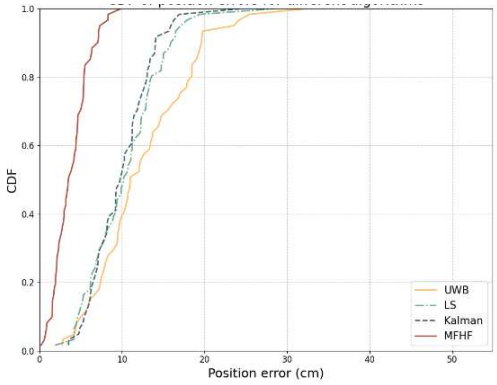


Fig. 44. CDF of positioning error measured for each method in the corridor scene.

7.217 cm, respectively. The MAE of the original UWB positioning, LS, Kalman, and MFHF were 19.213, 13.597, 11.520, and 6.720 cm, respectively. Due to the complexity of the laboratory environment, the UWB positioning trajectories exhibit significant fluctuations. In contrast, the proposed method can better alleviate the accuracy fluctuations caused by NLOS scene switching. All three methods offer relatively good error correction compared to the original UWB ranging. Compared with the original UWB ranging, the performance of the MFHF method in RMSE and MAE has been improved by an average of 66.43% and 65.02%, respectively. Compared with the LS method, the average performance of the MFHF method still has improvements of 53.04% and 50.68%, respectively. The RMSE and MAE of the Kalman method have performance gaps of 43.38% and 41.67%, respectively, compared with the MFHF method. Overall, the MFHF method demonstrates robust performance across all trajectories, significantly improving UWB positioning accuracy. This is because in the process of feature extraction, various channel conditions and UWB original ranging information are fully considered, reducing the positioning error caused by NLOS. These results indicate that the proposed channel multiclassification fusion method based on UWB ranging information substantially enhances positioning accuracy. In terms of cdf, the cdf of the positioning errors of the three positioning trajectories also indicates that MFHF has better positioning performance than other methods. Therefore, it can be concluded that the designed system can alleviate the NLOS effect of the UWB system and reduce the UWB positioning error caused by NLOS.

TABLE VII
MAE OF POSITIONING ERRORS FOR EACH METHOD IN THREE SCENARIOS

Method	Underground parking lot [cm]	Laboratory [cm]	Corridor [cm]	Average [cm]
UWB	19.47	25.55	12.62	19.213
LS	14.75	15.53	10.51	13.597
Kalman	11.68	12.85	10.03	11.520
MFHF	7.21	8.06	4.89	6.720

VI. CONCLUSION

To address performance degradation caused by NLOS interference in UWB positioning systems within complex indoor environments, this study proposes an NLOS identification and suppression scheme based on an MFHF convolutional network. A series of experiments evaluates the MFHF model. The model enhances NLOS identification and error suppression by fusing three feature types, UWB physical layer parameters, GAF-generated CIR image features, and CNN-extracted time series features. Hybrid fusion enables dynamic weighting of cross-modal features. Performance evaluations compared with traditional machine learning, deep learning networks, and feature fusion methods demonstrate the MFHF model's significant advantages in key metrics. In the NLOS binary classification task, MFHF achieved over 95% in precision, recall, accuracy, and $F1$ score, substantially outperforming traditional methods. For multiclass scene classification, MFHF demonstrates excellent recognition of different NLOS scenes. Compared with the top-performing CNN, it improves accuracy by 6.42%, precision by 4.58%, recall by 8.94%, and $F1$ score by 6.57%. In error distribution analysis, the MFHF algorithm shows strong stability and minimal localization errors: most errors are below 5 cm, with the highest precision. For NLOS error regression, MFHF reduces MAE to 0.1286 m and RMSE to 0.2128 m. Its R^2 reaches 0.9493, outperforming DNN, SVM, and CNN by 47.14%, 46.13%, and 35.15% in MAE reduction, and 36.08%, 35.46%, and 24.16% in RMSE reduction, respectively. Environmental analysis shows better model performance in parking lot scenarios than in laboratory environments. The laboratory's dense obstacles generate ambient noise, reducing classification performance. CNN accuracy drops by 5%, and MFHF by 4%—indicating noise's impact on feature extraction. Regarding model complexity, MFHF has a three-layer architecture. While its training time was significantly longer than CNN's, its testing complexity remains low, with negligible differences in test times among SVM, CNN, DNN, and MFHF.

In short, the MFHF model achieves satisfactory results in NLOS recognition and error mitigation. In addition, we aim to explore how the results of UWB signal propagation channel identification can be more effectively applied to UWB indoor localization in harsh environments.

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